CentraleSupélec

Order–disorder transition of β-brass (CuZn)

Metropolis Monte Carlo on a rigid lattice

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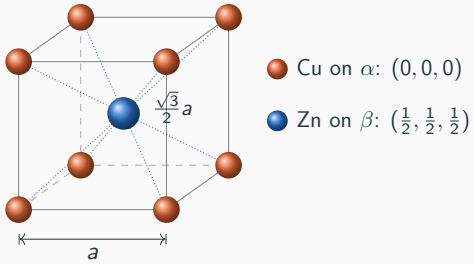
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CentraleSupélec — Modélisation et simulation numérique

The question

- β-brass: CuZn, 50 % Zn, bcc lattice
- Low T: ordered B2, Cu on corners (α), Zn on centres (β)
- High T: Cu and Zn mixed at random
- Transition at $T_c \approx 740\,\mathrm{K}$ [1, 2]

Goal: find T_c and see how order disappears, with our own Monte Carlo code



Model: rigid lattice + pair energies

- Box of $6 \times 6 \times 6$ cells: 432 sites, periodic boundaries
- Atoms never leave their site: only who sits where changes
- Energy = sum over pairs

$$E = \sum_{\text{1st}} J_{\sigma_i \sigma_j} + \sum_{\text{2nd}} I_{\sigma_i \sigma_j}$$

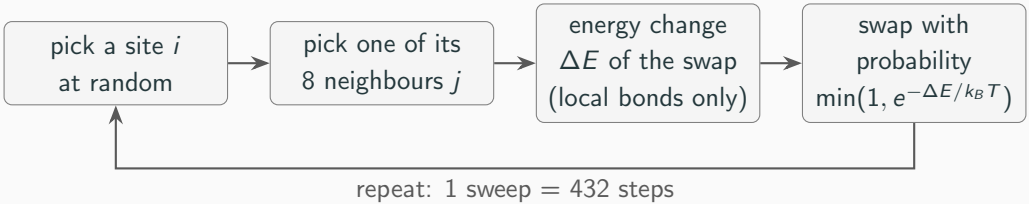
- 8 first neighbours (other sublattice), 6 second neighbours (same sublattice)

eV	Cu–Cu	Zn–Zn	Cu–Zn
J (1st)	-0.873	-0.338	-0.621
I (2nd)	0	0	$+0.005$

- 1st shell prefers Cu–Zn pairs
- 2nd shell prefers same-species pairs

$\Rightarrow$ both favour B2

Monte Carlo step



- Swaps keep 216 Cu + 216 Zn (Kawasaki dynamics [3]), Metropolis rule [4]
- ΔE uses 26 bonds instead of 3024: 7 µs per step in Python

What we measure

- Long-range order (do Cu atoms prefer sublattice α?)

$$\eta = \frac{N_{\mathrm{Cu}}^{(\alpha)} - N_{\mathrm{Cu}}^{(\beta)}}{N_{\mathrm{Cu}}^{(\alpha)} + N_{\mathrm{Cu}}^{(\beta)}} \qquad \begin{cases} 1 & \text{perfect B2} \\ 0 & \text{random} \end{cases} \qquad \text{we average } |\eta|$$

- Short-range order (Cowley [5]): are the neighbours of a Cu more often Zn than at random?

$$\alpha_n = 1 - \frac{p_n(\mathrm{Zn}\,|\,\mathrm{Cu})}{1/2} \qquad \text{B2: } \alpha_1 = -1,\ \alpha_2 = +1 \qquad \text{random: } 0$$

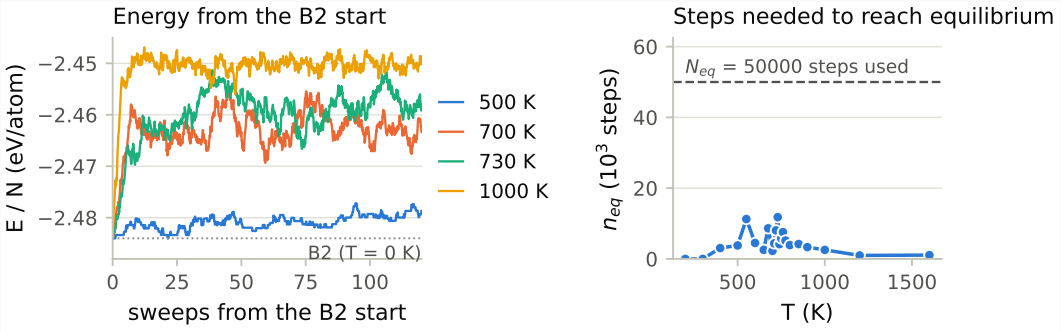
- Fluctuations peak at the transition: heat capacity $C \propto \langle E^2 \rangle - \langle E \rangle^2$, susceptibility $\chi \propto \langle \eta^2 \rangle - \langle |\eta| \rangle^2$

Is the code right?

Test	Expected	Obtained
Neighbour tables (8 + 6, symmetric)	yes	yes
Perfect B2: E/N	$4J_{\mathrm{CuZn}} = -2.484$ eV	-2.4840 eV
Perfect B2: η, α_1, α_2	$1, -1, +1$	$1, -1, +1$
Random lattices: α_1, α_2	0, 0	-0.002, -0.004
Random lattices: $\langle\lvert\eta\rvert\rangle$	0.038 (finite size)	0.038
Energy updated step by step vs recomputed	0	4×10^{-12} eV

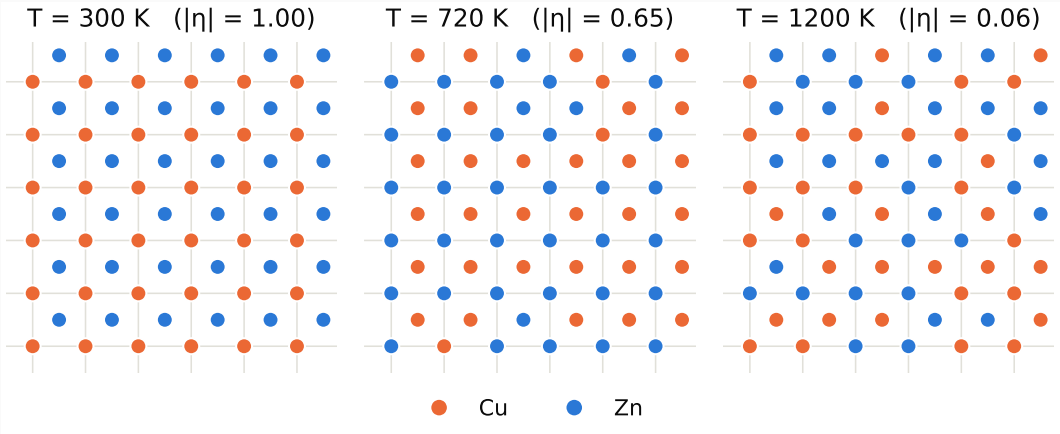
- Cohesive energy at 0 K: 2.484 eV/atom
- Same model for pure Cu and Zn: 3.49 and 1.35 eV/atom = experimental values [6]

How long before equilibrium?



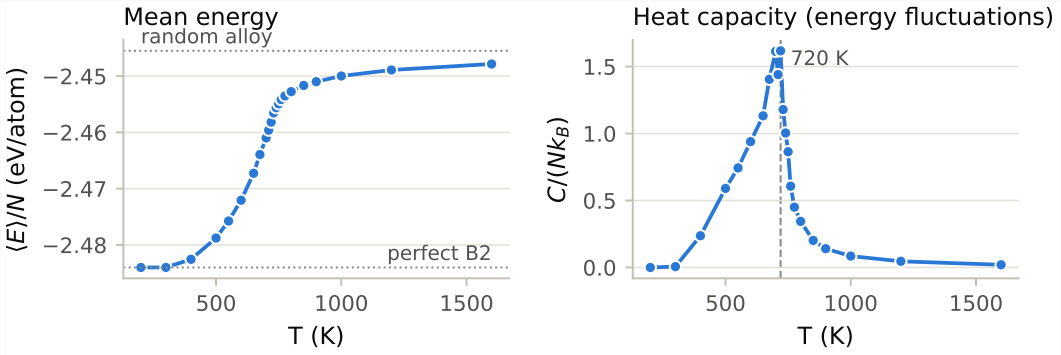
- Every run starts from perfect B2 and must “forget” it
- Slowest: about 12 000 steps ⇒ we discard 50 000 steps, then measure 4000 sweeps

What the crystal looks like



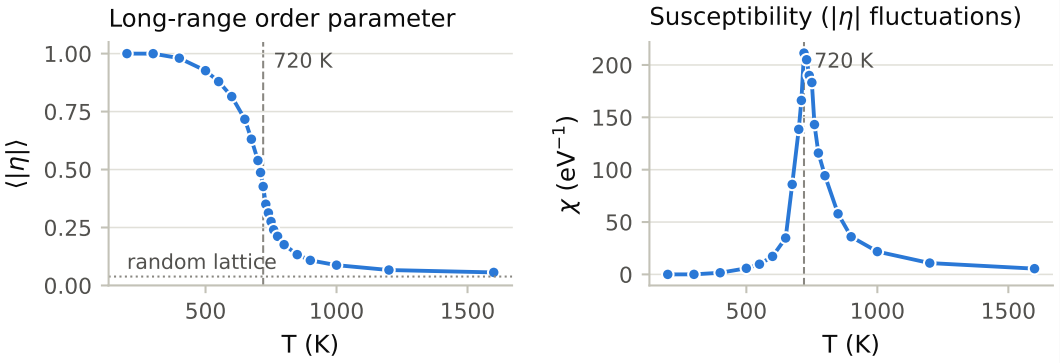
One layer of the box: α sites on the grid nodes, β sites in the squares.

Energy and heat capacity



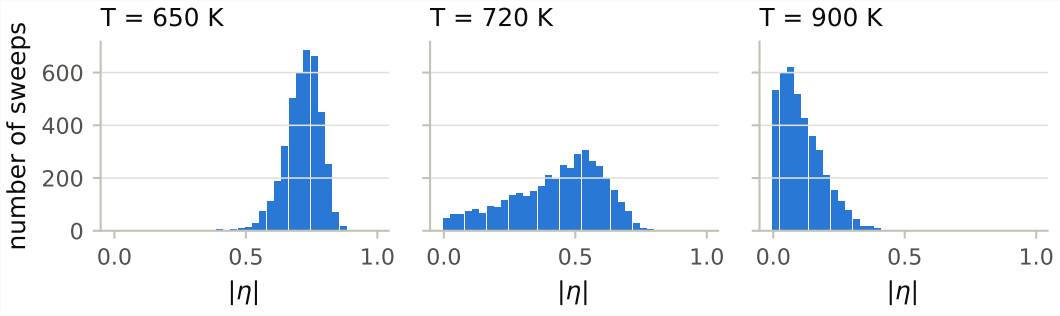
- Energy rises steeply between 600 and 800 K
- Heat capacity peak at 720 K (flat top 700–720 K)

Long-range order



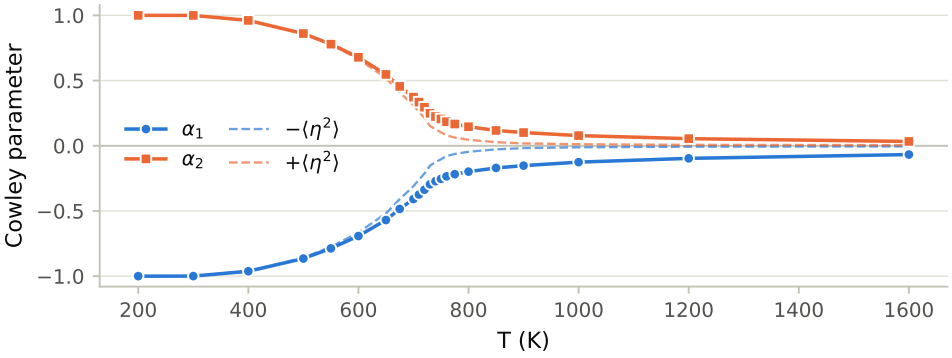
- $\langle|\eta|\rangle$ drops fastest at 720 K, susceptibility peak at 720 K
- Small box: drop is rounded, $|\eta|$ never below the random value 0.04

Near T_c the system hesitates



- Below: always ordered Above: always disordered
- Near T_c: goes back and forth $\Rightarrow$ large fluctuations $\Rightarrow$ peaks of C and χ

Short-range order survives above T_c



- Dashed: what long-range order alone gives ($\alpha_1 = -\eta^2$, $\alpha_2 = +\eta^2$)
- Above T_c: $\eta \approx 0$ but $\alpha_1 < 0 \Rightarrow$ Cu still prefers Zn neighbours

Transition temperature

Method	T_c (K)
Peak of C	720
Peak of χ	720
Steepest drop of $\lvert\eta\rvert$	720
Mean field (Bragg–Williams)	894
Experiment	≈ 740

- Monte Carlo: close to experiment
- Mean field [7]: $k_B T_c = 2\,(E_{\text{random}} - E_{\text{B2}})$, 170 K too high
- Why? Mean field ignores short-range order, which stabilises the disordered phase

Limits and conclusion

Limits

- Small box (432 atoms): rounded transition, broad peaks
- Rigid lattice, pair energies: no vibrations, no size mismatch
- Near T_c: slow, correlated sampling $\Rightarrow$ noisy C and χ

Conclusion

- Tested Metropolis code; cohesive energy 2.484 eV/atom
- Order–disorder transition at $T_c \approx 720$ K (experiment ≈ 740 K)
- Short-range order above T_c explains why mean field overestimates T_c
- Next step: several box sizes to extrapolate T_c to an infinite crystal [8]

References i

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[3] K. Kawasaki. **"Diffusion Constants near the Critical Point for Time-Dependent Ising Models. I".** In: *Physical Review* 145.1 (1966), pp. 224–230.

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[6] C. Kittel. ***Introduction to Solid State Physics.*** 8th ed. Hoboken, NJ: John Wiley & Sons, 2005.

References ii

[7] W. L. Bragg and E. J. Williams. **"The effect of thermal agitation on atomic arrangement in alloys".** In: *Proceedings of the Royal Society of London. Series A* 145.855 (1934), pp. 699–730.

[8] D. P. Landau and K. Binder. ***A Guide to Monte Carlo Simulations in Statistical Physics*.** 4th ed. Cambridge: Cambridge University Press, 2014.

[9] ***Study of the order-disorder transition of the CuZn compound by the Monte Carlo method on a rigid lattice*.** Project description, course *Modélisation et simulation numérique*, chapter 2. 2026.

[10] K. Binder and D. W. Heermann. ***Monte Carlo Simulation in Statistical Physics: An Introduction*.** 6th ed. Graduate Texts in Physics. Cham: Springer, 2019.

[11] S. C. Moss. **"X-Ray Measurement of Short-Range Order in Cu_3Au".** In: *Journal of Applied Physics* 35.12 (1964), pp. 3547–3553.

[12] S. N. Khan and M. Eisenbach. **"Density-functional Monte-Carlo simulation of CuZn order-disorder transition".** In: *Physical Review B* 93 (2016), p. 024203.

Backup: simulation conditions

Lattice	bcc, 6^3 cells, 432 sites, periodic boundaries
Start	perfect B2 at every temperature
Moves	swap with a random first neighbour, Metropolis
Equilibration	50 000 steps (116 sweeps), discarded
Measurement	4000 sweeps, one measurement per sweep
Temperatures	22 values, 200–1600 K, step 10 K near T_c
Random numbers	Python `random` (Mersenne Twister), fixed seeds
Cost	6.7 µs per step, 6 min for the whole study

Backup: why $\alpha_1 = -\eta^2$ with long-range order only

- Fraction of Cu on α: $\frac{1+\eta}{2}$ on β: $\frac{1-\eta}{2}$
- A Cu on α has its 1st neighbours on β, Zn with probability $\frac{1+\eta}{2}$
- A Cu on β has its 1st neighbours on α, Zn with probability $\frac{1-\eta}{2}$

$$p_1(\mathrm{Zn}|\mathrm{Cu}) = \left(\frac{1+\eta}{2}\right)^2 + \left(\frac{1-\eta}{2}\right)^2 = \frac{1+\eta^2}{2} \;\Rightarrow\; \alpha_1 = 1 - 2p_1 = -\eta^2$$

- Same reasoning on the same sublattice: $\alpha_2 = +\eta^2$